



Student Number	
Mark / 35	

Chemistry

Preliminary Course

Final Examination • 2005

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided at the back of this paper and may be removed for convenience
- Write your Student Number at the top of this page

Total Marks – 35

Part A – 9 marks

- Attempt Questions 1 – 9
- Allow about 10 minutes for this part

Part B – 26 marks

- Attempt Questions 10 – 19
- Allow about 40 minutes for this part

Allow about 10 minutes for this part

A  B  C  D 

correct

Answer Box for Questions 1 – 9				
1	A ☐	B ☐	C ☐	D ☐
2	A ☐	B ☐	C ☐	D ☐
3	A ☐	B ☐	C ☐	D ☐
4	A ☐	B ☐	C ☐	D ☐
5	A ☐	B ☐	C ☐	D ☐
6	A ☐	B ☐	C ☐	D ☐
7	A ☐	B ☐	C ☐	D ☐
8	A ☐	B ☐	C ☐	D ☐
9	A ☐	B ☐	C ☐	D ☐

1 Study the reaction... $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$

Which of the statements cannot apply to this reaction?

- (A) One gram of H_2 reacts with one gram of Cl_2 to produce 2 grams of HCl .
- (B) One molecule of H_2 reacts with one molecule of Cl_2 to produce 2 molecules of HCl .
- (C) One litre of H_2 reacts with one litre of Cl_2 to produce 2 litres of HCl at constant conditions.
- (D) One mole of H_2 reacts with one mole of Cl_2 to produce 2 moles of HCl .

2 Which of the production sequences shows the extraction of copper from its ore to 99.9% pure copper?

- (A) froth flotation $\rightarrow$ crushing and grinding $\rightarrow$ smelting $\rightarrow$ electrolysis
- (B) crushing and grinding $\rightarrow$ froth flotation $\rightarrow$ smelting $\rightarrow$ electrolysis
- (C) crushing and grinding $\rightarrow$ smelting $\rightarrow$ froth flotation $\rightarrow$ electrolysis
- (D) smelting $\rightarrow$ crushing and grinding $\rightarrow$ froth flotation $\rightarrow$ electrolysis

3 Which of the following may **not** be a consequence of thermal pollution in water?

- (A) Reduction in dissolved oxygen.
- (B) Disruption of aquatic organisms breeding cycles.
- (C) Out of season migration of aquatic fauna.
- (D) Decrease in salt concentration.

4 At 100 kPa and 25°C, 4 litres of oxygen gas contain 1×10^{21} molecules.
Which of these gas samples also contains 1×10^{21} molecules under the same conditions?

- (A) 1 L of NH_3
- (B) 2 L of Cl_2
- (C) 4 L of CO_2
- (D) 8 L of He

5 To which reaction can Gay–Lussac’s law be applied?

- (A) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$
(B) $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$
(C) $2\text{H}_2\text{O}(\text{l}) + 2\text{Na}(\text{s}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$
(D) $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

6 Which property accounts for the moderate viscosity of water?

- (A) Specific heat capacity
(B) Hydrogen bonding
(C) Density
(D) Boiling point

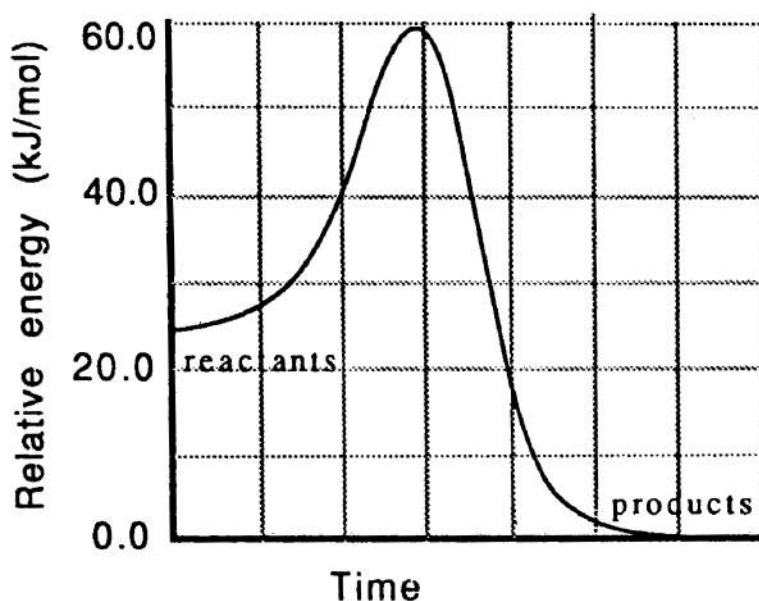
7 Metals X, M, Z and D are all very useful metals. The table shows selected properties of the metals...

<i>Property</i>	<i>X</i>	<i>M</i>	<i>Z</i>	<i>D</i>
<i>Ionisation energy (kJ mol⁻¹)</i>	584	751	896	766
<i>Percentage Abundance in Earth’s Crust</i>	8	0.07	0.00001	0.07

Use the data to list the metals according to increasing market price.

- (A) $X < M < D < Z$
(B) $D < X < Z < M$
(C) $X < Z < M < D$
(D) $Z < D < M < X$

Answer Questions 8 and 9 using this reaction pathway diagram...



- 8 Which statement correctly describes this reaction?
- (A) The reaction is endothermic and the surroundings will become cooler.
 - (B) The reaction is exothermic and the surroundings will become cooler.
 - (C) The reaction is endothermic and the surroundings will become warmer.
 - (D) The reaction is exothermic and the surroundings will become warmer.
- 9 Which statement is true if a catalyst is added to the system?
- (A) ΔH remains constant and the activation energy increases.
 - (B) Both ΔH and activation energy decrease.
 - (C) ΔH remains constant and the activation energy decreases.
 - (D) Both ΔH and activation energy remain constant.

Part B – 26 marks

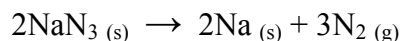
Attempt Questions 10 – 19

Allow about 40 minutes for this part

► *Show all relevant working in questions involving calculations.*

Question 10 (3 marks)

Air bags have saved thousands of lives and are now commonly fitted in new cars. An air bag inflates very rapidly by producing nitrogen gas from the decomposition of 100 grams of sodium azide...



- (a) Calculate the moles of sodium azide originally in the air bag. **(1 mark)**

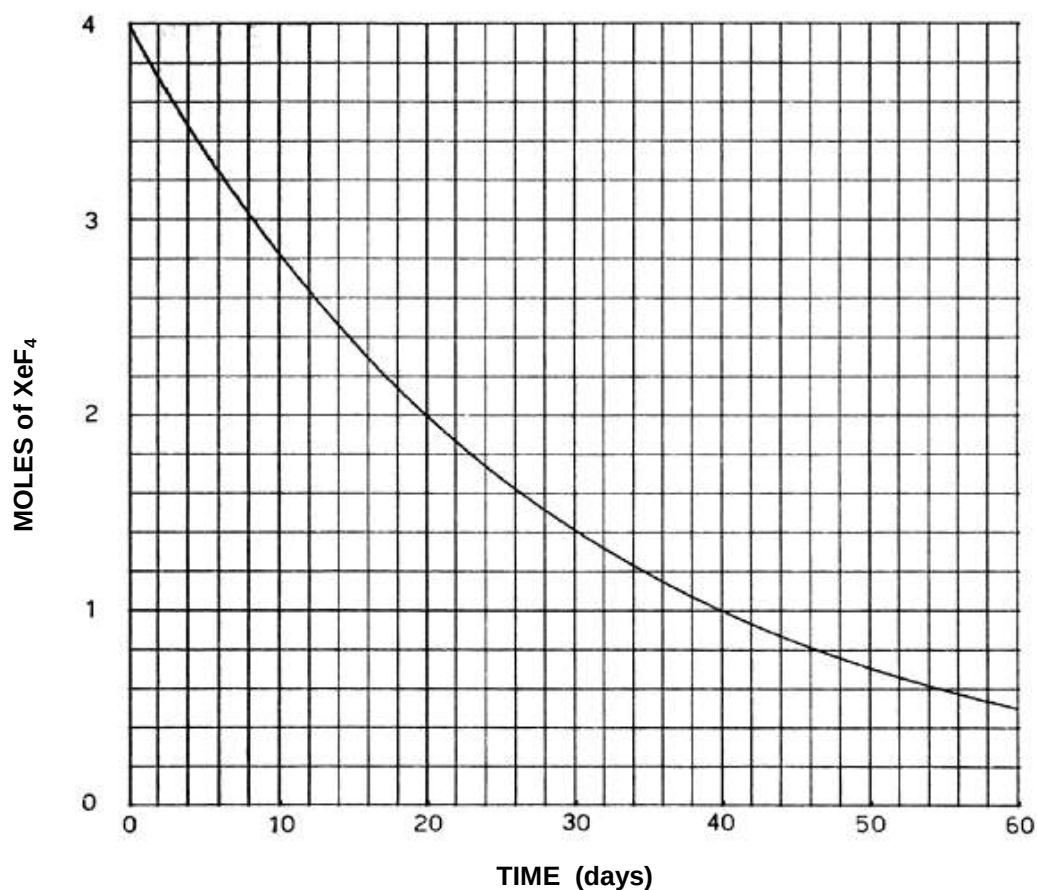
- (b) Calculate the moles of nitrogen produced and the resultant volume at 100 kPa and 25°C. **(2 marks)**

Question 11 (3 marks)

Xenon tetrafluoride is an unstable compound which self-decomposes...



The graph shows the decomposition of a pure, 4 mole sample of XeF_4 over a period of 60 days.



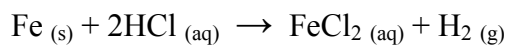
- (a) Calculate the number of moles of XeF_4 which has decomposed after 16 days. **(1 mark)**

- (b) On which day are there equal moles of reactant and products present? **(1 mark)**

- (c) Calculate the mass of F_2 present at 40 days. **(1 mark)**

Question 12 (2 marks)

Iron will react with hydrochloric acid and appear to “dissolve” ...

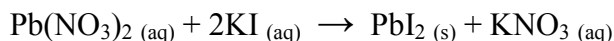


Ken Chemiski places a 2.51 g iron nail in a beaker and prepares some dilute, 1.00 mol L^{-1} HCl.

Calculate the volume of acid required to fully react with the nail.

Question 13 (3 marks)

Héloïse prepares some beautiful golden lead(II) iodide crystals by this precipitation reaction...



She reacts 25.0 mL of 0.100 mol L^{-1} potassium iodide with excess lead(II) nitrate solution.

(a) Calculate the moles of lead(II) nitrate which reacted. **(2 marks)**

(b) Calculate the mass of PbI_2 produced. **(1 mark)**

Question 14 (3 marks)

In your studies, you performed a practical investigation to observe the effect of temperature on reaction rate. Describe your experiment, including the observed results.

Question 15 (2 marks)

- (a) Identify one pollutant produced by the incomplete combustion of an organic compound. **(1 mark)**

- (b) Write a chemical equation to show the incomplete combustion of butane (C_4H_{10}). **(1 mark)**

Question 16 (2 marks)

The diagram shows liquid methanol, CH_3OH , being poured into a beaker of water.



Use the symbols, $\text{H}-\text{O}-\text{H}$ for water and $\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{H} \\ | \\ \text{O}-\text{H} \end{array}$ for methanol to draw a diagram that illustrates the strongest intermolecular forces in the solution.

Question 17 (3 marks)

Outline the method required to prepare a 250 mL solution of 0.102 mol L^{-1} strontium chloride.

► *Write your answer in numbered sequential steps (1, 2, 3...).*

Question 18 (2 marks)

Tincture of iodine, a common antiseptic, is a 2% (w/w) solution of iodine in ethanol.

- (a) Calculate the mass of iodine crystals is required to prepare a 250 g sample of tincture of iodine? **(1 mark)**

- (b) Calculate the mass of ethanol is required? **(1 mark)**

Question 19 (3 marks)

In an experiment to measure the density of water at 28°C, an empty 100 mL measuring cylinder was found to have a mass of 150.5 grams. When 80.3 mL of water was poured into the cylinder, the mass of the cylinder and contents was 230.5 grams.

- (a) Calculate the density of water at 28°C based on the data. **(1 mark)**

- (b) When the measuring cylinder and water was sealed and kept inside a freezer overnight at – 10°C, the volume reading increased to 85.9 mL. Explain this observation. **(1 mark)**

- (c) When water freezes it expands. Why is this process important in Nature? **(1 mark)**

HIGHER SCHOOL CERTIFICATE EXAMINATION
Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

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² He	4.003
Helium	
¹⁰ Ne	20.18
Neon	
¹⁸ Ar	39.95
Argon	
³⁶ Kr	83.80
Krypton	
⁵⁴ Xe	131.3
Xenon	
⁸⁶ Rn	222.01
Radon	



Chem 11
Final Exam – 2005



ANSWERS

1	A
2	B
3	D

4	C
5	A
6	B

7	A
8	D
9	C

- 10 (a) $n = m \div M = 100 \text{ g} \div 65.02 \text{ g mol}^{-1} = \mathbf{1.54 \text{ mol}}$ (1 mark)
- (b) $\text{moles N}_2 = 1\frac{1}{2} \text{ moles NaN}_3 = 1\frac{1}{2} \times 1.54 \text{ mol} = \mathbf{2.31 \text{ mol}}$ (1 mark)
- $\text{volume N}_2 = n \times 24.79 \text{ L mol}^{-1} = 2.31 \text{ mol} \times 24.79 \text{ L mol}^{-1} = \mathbf{57.2 \text{ L}}$ (1 mark)
- 11 (a) 4 moles initially – 2.3 moles remaining @ Day 16 = **1.7 moles** decomposed (1 mark)
- (b) At **Day 8**, 3 moles of XeF_4 remain and 1 mole of XeF_4 has decomposed yielding 1 mole of Xe and 2 moles of F_2 . (1 mark)
- (c) At Day 40, 3 moles of XeF_4 has decomposed, yielding 6 moles of F_2 .
- $n = m \div M$; $m = n \times M = 6 \text{ mol} \times 38.00 \text{ g mol}^{-1} = \mathbf{228 \text{ g}}$ (1 mark)
- 12 $\text{moles Fe} = m \div M = 2.51 \text{ g} \div 55.85 \text{ g mol}^{-1} = 0.0449 \text{ mol}$
- $\text{moles HCl} = 2 \times \text{moles Fe} = 2 \times 0.0449 \text{ mol} = 0.0899 \text{ mol}$ (1 mark)
- $\text{volume HCl} = n \div c = 0.0899 \text{ mol} \div 1.00 \text{ mol L}^{-1} = \mathbf{0.0899 \text{ L}}$ (1 mark)
- 13 (a) $\text{moles Pb(NO}_3)_2 = \frac{1}{2} \times \text{moles KI} = \frac{1}{2} \times c \times V = \frac{1}{2} \times 0.100 \text{ mol L}^{-1} \times 0.0250 \text{ L}$
- $\text{moles Pb(NO}_3)_2 = \mathbf{0.00125 \text{ mol}}$ (1 mark + 1 mark)
- (b) $\text{moles PbI}_2 = \frac{1}{2} \text{ moles KI} = \frac{1}{2} \times 0.00250 \text{ mol} = 0.00125 \text{ mol}$
- $\text{mass PbI}_2 = n \times M = 0.00125 \text{ mol} \times 461 \text{ g mol}^{-1} = \mathbf{0.576 \text{ g}}$ (1 mark)

- 14 **Sodium thiosulfate was added to hydrochloric acid in a conical flask at various temperatures. The chemicals react to form a precipitate. The time was recorded for how long it took for the precipitate to obscure a cross drawn on the bottom of the conical flask. As the flask was heated, the time taken for the precipitate to obscure the cross became shorter.**

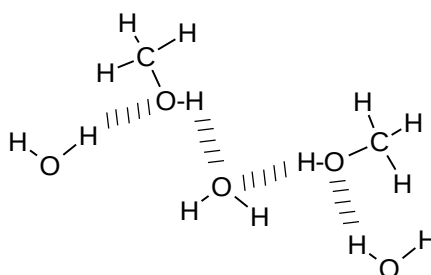
- Detailed description and observed results. 3 marks
 Detailed description of valid experiment. 2 marks
 Simple description of valid experiment. 1 mark

- 15 (a) **Carbon or carbon monoxide or carbon dioxide (1 mark)**

- (b) e.g. $\text{C}_4\text{H}_{10}(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow \text{C}(\text{s}) + \text{CO}(\text{g}) + 2\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l})$ (1 mark)

- No states required.

- 16 |||| indicates hydrogen bonding



Hydrogen bonding shown using dashes or any suitable representation (but not solid lines) between correct atoms. (1 mark)

Written label indicating the hydrogen bond location. (1 mark)

1. Calculate the mass of strontium chloride required to be weighed.

$$\text{Mass of strontium chloride} = 0.250 \text{ L} \times 0.102 \text{ mol L}^{-1} \times [87.62 \times 2(35.45)] \text{ g mol}^{-1} = 4.04 \text{ g}$$

2. Weigh the required quantity and dissolve in the minimum amount of water.
3. Quantitatively transfer the solution to a 250 mL volumetric flask and add enough water until the lower meniscus is just touching the fill line.
4. With the stopper on and held firmly in place, invert the volumetric flask several times to mix the solution.

► Steps 1 and 2 are credited with 1 mark. Numbers 3 and 4 are 1 mark each.

- 18 (a) mass of iodine = $0.02 \times 250 = 5 \text{ g}$ (1 mark)

- (b) mass of ethanol = mass of solution – mass of solute = 245 g (1 mark)

- 19 (a) Density = mass H₂O ÷ volume H₂O = $(230.5 \text{ g} - 150.5 \text{ g}) \div 80.3 \text{ mL} = 0.996 \text{ g mL}^{-1}$

- (b) At -10°C , water freezes to a structure where each molecule hydrogen bonds with four other molecules (tetrahedrally configured) creating a regular open structure which occupies more space, hence the greater volume and lower density. (1 mark)

- (c) Ice forms on the surface of a lake (etc.) instead of the bottom allowing life to exist under the ice during the winter.

OR

Ice is an agent of physical weathering for rocks. Successive freezing and thawing can crack open the surface of hard rock forming a component of soil. (1 mark)